

ÔN TẬP CHƯƠNG 3 CĂN THỨC

Bài 1. Rút gọn biểu thức:

1) $4\sqrt{5} + \sqrt{45} - 3\sqrt{20}$

3) $\sqrt{12} + \sqrt{27} - \sqrt{48}$

5) $5\sqrt{12} + 2\sqrt{48} - 3\sqrt{300}$

7) $\sqrt{208} - 2\sqrt{117} + \frac{3}{2}\sqrt{52}$

9) $\frac{1}{2}\sqrt{48} - 5\sqrt{27} + 2\sqrt{147} - \sqrt{108}$

2) $3\sqrt{20} + \sqrt{75} - 2\sqrt{45}$

4) $\sqrt{75} - \sqrt{108} + \sqrt{27}$

6) $\sqrt{12} - 4\sqrt{48} + \sqrt{243} + 2\sqrt{147}$

8) $\sqrt{243} + 0,5\sqrt{12} - \sqrt{75} + \frac{2}{7}\sqrt{147}$

10) $\sqrt{75} - 3\sqrt{2} + \sqrt{27} - \sqrt{192}$

Bài 2. Rút gọn biểu thức sau:

1) $\sqrt{16 - 8\sqrt{3}} - \sqrt{(3 - 2\sqrt{3})^2}$

3) $\sqrt{49 + 20\sqrt{6}} - \sqrt{(3 - 2\sqrt{6})^2}$

5) $\sqrt{9 - 4\sqrt{5}} \cdot \sqrt{(2 + \sqrt{5})^2}$

7) $(\sqrt{14} + \sqrt{6}) \cdot \sqrt{5 - \sqrt{21}}$

9) $\sqrt{14 - 5\sqrt{3}}(5\sqrt{2} + \sqrt{6})$

11) $(\sqrt{3} - \sqrt{5} + \sqrt{2 + \sqrt{3}})(\sqrt{10} - \sqrt{6})$

13) $\sqrt{4 + \sqrt{6 - 2\sqrt{5}}}(\sqrt{10} - \sqrt{2})$

15) $(\sqrt{10} + \sqrt{6})\sqrt{4 + \sqrt{15}}(4 - \sqrt{15})$

17) $\sqrt{(4 + \sqrt{3})\sqrt{19 - 8\sqrt{3}}} + 3$

19) $(1 + \sqrt{3})(2 - \sqrt{3}) + \sqrt{9 - 2\sqrt{13 + 4\sqrt{3}}}$

2) $\sqrt{14 + 6\sqrt{5}} - \sqrt{(\sqrt{5} - 2)^2}$

4) $\sqrt{(2\sqrt{2} - 3)^2} - \sqrt{11 - 6\sqrt{2}}$

6) $\sqrt{(\sqrt{3} - 2)^2} \cdot \sqrt{7 + 4\sqrt{3}}$

8) $(\sqrt{10} + \sqrt{6}) \cdot \sqrt{4 - \sqrt{15}}$

10) $\sqrt{2 - \sqrt{3}} \cdot (\sqrt{2} + \sqrt{6})$

12) $(\sqrt{11 - 6\sqrt{2}} - \sqrt{15 - 6\sqrt{6}})\sqrt{2 + \sqrt{3}}$

14) $\sqrt{2 - \sqrt{6 - 2\sqrt{5}}} \cdot (\sqrt{10} + \sqrt{2})$

16) $(\sqrt{10} + \sqrt{2})\sqrt{3 + \sqrt{5}}(3 - \sqrt{5})$

18) $(4 - \sqrt{15})(\sqrt{2 - \sqrt{3}} + \sqrt{3 + \sqrt{5}})^2$

20) $\sqrt{\sqrt{5} - \sqrt{3} - \sqrt{29 - 12\sqrt{5}}}$

Đs: 1) 1; 2) $2\sqrt{5}$; 3) 8; 4) $-\sqrt{2}$; 5) 1; 6) 1; 7) 4; 8) 2; 9) 22; 10) 2; 11) 2;

12) 2; 13) 4; 14) 4; 15) 2; 16) 8; 17) 4; 18) 1; 19) 1; 20) 1.

Bài 3. Rút gọn biểu thức sau:

1) $\frac{2\sqrt{3} - 3\sqrt{2}}{\sqrt{3} - \sqrt{2}} + 3\sqrt{\frac{2}{3}} + \frac{4}{2 - \sqrt{6}}$

3) $\frac{-2}{2\sqrt{5} + 3} + \frac{1}{2\sqrt{5} - 3} + \frac{2\sqrt{5}}{11}$

5) $\sqrt{13 - 4\sqrt{3}} - \sqrt{\frac{2}{2 + \sqrt{3}}}$

2) $\frac{2}{3 - \sqrt{10}} - \frac{36}{4 + \sqrt{10}} - \frac{40}{\sqrt{10}}$

4) $\frac{-1}{3 + 2\sqrt{3}} + \frac{1}{3 - 2\sqrt{3}} + \frac{-1}{\sqrt{3}}$

6) $\frac{6}{\sqrt{5} + 1} + \sqrt{\frac{2}{7 + 3\sqrt{5}}}$

$$7) \sqrt{\frac{2}{2-\sqrt{3}}} + \sqrt{4-2\sqrt{3}} - \sqrt{27}$$

$$9) \frac{4}{\sqrt{5}-1} + \sqrt{\frac{8}{7+3\sqrt{5}}}$$

$$11) \left(\frac{\sqrt{6}-\sqrt{2}}{\sqrt{3}-1} + \frac{1}{\sqrt{3}+\sqrt{2}} \right) : \frac{1}{2\sqrt{3}}$$

$$13) \frac{5-\sqrt{15}}{\sqrt{5}-\sqrt{3}} - \frac{2\sqrt{5}-6}{\sqrt{14-2\sqrt{45}}}$$

$$15) \frac{5-\sqrt{5}}{\sqrt{5}-1} - \sqrt{\frac{\sqrt{5}+2}{\sqrt{5}-2}}$$

$$17) \frac{(\sqrt{10}+\sqrt{6})\sqrt{4-\sqrt{15}}}{8-2\sqrt{15}} - \sqrt{15}$$

$$19) \frac{3+\sqrt{5}}{5-\sqrt{9+4\sqrt{5}}} + \frac{3-\sqrt{5}}{4+\sqrt{6-2\sqrt{5}}}$$

$$8) \frac{-2}{\sqrt{5}+1} + \sqrt{\frac{2}{3-\sqrt{5}}} - \frac{2}{\sqrt{3}-1}$$

$$10) \sqrt{\frac{2}{7+3\sqrt{5}}} + \frac{2}{3-\sqrt{5}} - \frac{2+3\sqrt{2}}{\sqrt{2}}$$

$$12) \left(\frac{\sqrt{6}+\sqrt{2}}{1+\sqrt{3}} - \frac{5}{\sqrt{5}} \right) : \frac{\sqrt{5}+\sqrt{2}}{3}$$

$$14) \frac{12}{4-\sqrt{10}} - 6\sqrt{\frac{5}{2}} + \frac{5\sqrt{2}+\sqrt{10}}{\sqrt{5}+1}$$

$$16) \sqrt{14+6\sqrt{5}} - \sqrt{\frac{\sqrt{5}-2}{\sqrt{5}+2}}$$

$$18) \sqrt{\frac{\sqrt{3}+4}{5-2\sqrt{3}}} \cdot (\sqrt{6}-\sqrt{2})$$

$$20) \sqrt{\frac{2\sqrt{10}+\sqrt{30}-2\sqrt{2}-\sqrt{6}}{2\sqrt{10}-2\sqrt{2}}} + \frac{1}{\sqrt{3}+1}$$

Đs: 1) $-2\sqrt{6}-4$; 2) -30 ; 3) $\frac{9}{11}$; 4) $\frac{-5\sqrt{3}}{3}$; 5) $\sqrt{3}$; 6) $\sqrt{5}$; 7) $-\sqrt{3}$; 8) $-\sqrt{3}$;

9) 4 ; 10) $-\sqrt{2}$; 11) 6 ; 12) -1 ; 13) $\sqrt{5}+2$; 14) 8 ; 15) -2 ; 16) 5 ; 17) 4 ; 18) 2 ; 19) 7 ; 20) $\sqrt{3}$

Bài 4: Giải các phương trình sau:

$$1) \sqrt{4(1-2x)^2} = 6$$

$$3) \sqrt{36x^2-12x+1} = 3$$

$$5) \sqrt{4x^2-12x+9} + 5x = 4$$

$$7) \sqrt{x^2-5} = \sqrt{4x-9}$$

$$9) \sqrt{9x^2-12x+4} = \sqrt{4x^2-12x+9}$$

$$11) \sqrt{9x^2-3x+1} = 3x-2$$

$$13) 2\sqrt{12-4x} + \frac{3}{4}\sqrt{48-16x} - 3\sqrt{\frac{75-25x}{9}} = 6$$

$$15) \sqrt{4x+20} + \frac{5}{4}\sqrt{\frac{16x+80}{25}} - \frac{2}{3}\sqrt{9x+45} = 4$$

$$17) \sqrt{4-8x} + \frac{6}{5}\sqrt{25-50x} - 49\sqrt{\frac{9-18x}{49}} + 26 = 0$$

$$2) \sqrt{9-2x} = \sqrt{x^2+9}$$

$$4) \sqrt{9x^2-6x+1} - 2 = 0$$

$$6) \sqrt{9x^2-12x+4} + 4x = 5$$

$$8) \sqrt{4x^2+x-3} = 2x$$

$$10) \sqrt{x^2-4x+4} = \sqrt{100x^2}$$

$$12) \sqrt{x^2-2} - x = 1$$

$$14) 5\sqrt{4x-16} - 2\sqrt{25x-100} + 14\sqrt{\frac{9x-36}{49}} = 42$$

$$16) \sqrt{4x^2+4} - 3\sqrt{\frac{x^2+1}{9}} = 2$$

$$18) \sqrt{9x^3-9} + 5\sqrt{\frac{x^3-1}{25}} = \sqrt{112}$$

Bài 5. Cho biểu thức $P = (2 + \sqrt{a}) \left(\frac{1}{\sqrt{a} + 2} - \frac{1}{2 - \sqrt{a}} + \frac{a}{4 - a} \right)$ với $a \geq 0, a \neq 4$.

- a) Rút gọn biểu thức P .
- b) Tính giá trị của P khi $a = 0,81$.
- c) Tìm giá trị của a để $P = 5$

Bài 6. Một chiếc thùng hình lập phương có chiều dài cạnh là $x(\text{cm})$.

- a) Viết công thức tính thể tích $V(\text{cm}^3)$ và tổng diện tích $S(\text{cm}^2)$ các mặt theo x .
- b) Viết công thức tính x theo S .
- c) Viết công thức tính V theo S . Tính V khi $S = 50 \text{ cm}^2$.